

Indices and derived variables

CFTR AlphaGenome splice benchmark — pre-registration supplementary file

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Summary

No composite indices, no weighted sums, no factor scores are used in this study. Every quantity used in the confirmatory analyses is either a direct measurement from a published source or a single-value output from a public in silico splice-prediction model, transformed only by the minimal deterministic rules given below. Each of the three predictor scores per variant is evaluated on its own; no ensemble, no downstream model, and no combined index is constructed.

1. AlphaGenome per-junction usage delta

For each variant v , let $u_{\text{REF}}(j, c)$ and $u_{\text{ALT}}(j, c)$ be the AlphaGenome-predicted junction usage at junction j in cell context c , for the reference and alternate alleles respectively. The variant-level score is:

$$S_{\text{AG}}(v) = |u_{\text{ALT}}(j^*, c^*) - u_{\text{REF}}(j^*, c^*)|$$

where j^* is the canonical splice junction closest to v (nearest by genomic distance; ties broken by upstream junction), and c^* is the pre-declared default lung / bronchial epithelial cell context. No aggregation across junctions and no aggregation across cell contexts.

2. SpliceAI Δ -max

For each variant v , let Δ_{d+} , Δ_{d-} , Δ_{a+} , Δ_{a-} be the four SpliceAI Δ -scores (donor gain, donor loss, acceptor gain, acceptor loss) at ± 500 nt from v , from the standard Illumina release on GRCh38. The variant-level score is:

$$S_{\text{SAI}}(v) = \max(\Delta_{d+}, \Delta_{d-}, \Delta_{a+}, \Delta_{a-})$$

taken as an absolute value (SpliceAI Δ -scores are non-negative in the standard release).

3. Pangolin usage delta

For each variant v , let $p_{\text{REF}}(s)$ and $p_{\text{ALT}}(s)$ be the Pangolin-predicted usage at reference splice site s . The variant-level score is:

$$S_{\text{PAN}}(v) = |p_{\text{ALT}}(s^*) - p_{\text{REF}}(s^*)|$$

where s^* is the reference splice site closest to v (nearest by genomic distance; ties broken by upstream site).

4. Stratum 1 ground-truth handling (multi-organ replicates)

For c.3717+12191C>T (Chiba-Falek 1999), five multi-organ % residual measurements exist. The pre-declared handling is to report the Stratum 1 Spearman correlation both ways:

Collapsed: the five multi-organ measurements are reduced to a single per-variant value equal to their arithmetic mean, so the variant contributes one data point.

Retained: each of the five multi-organ measurements is retained as a separate data point paired with the same per-variant predictor score.

If the collapsed and retained versions give materially divergent correlations (defined as absolute difference in Spearman $\rho \geq 0.2$), the collapsed version is treated as primary and the divergence is reported.

5. AUC-difference statistic (primary inferential)

For the paired ROC-AUC comparisons on Stratum 2, the derived quantity is the paired AUC difference:

$$D_{AG \text{ vs } SAI} = AUC(S_{AG}) - AUC(S_{SAI})$$

with the DeLong 95% confidence interval computed by the paired DeLong procedure (DeLong et al. 1988) as implemented in the R package pROC. The identical formula applies to the AlphaGenome vs Pangolin comparison, evaluated against the pre-declared non-inferiority margin of -0.05 on the AUC difference.

6. No derived indices

No predictor scores are combined, averaged, weighted, or transformed into a composite. No ensemble of AlphaGenome, SpliceAI, and Pangolin is constructed. No downstream model is trained on any predictor output. Each predictor is evaluated on its own, and the three per-predictor scores per variant are compared pairwise using the paired statistics declared in the Analysis Plan section of the OSF pre-registration.